Group RDMA : Reliable RDMA Multicast via End-Net Collaboration

—— A High-Performance Communication Infrastructure for HPC & AI

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Core Problem: The Trade-off Between Reliability and Efficiency

In HPC (e.g., HPL, WRF) and AI training, collective operations like MPI_Bcast are frequent and data-intensive. Existing solutions face a dilemma:

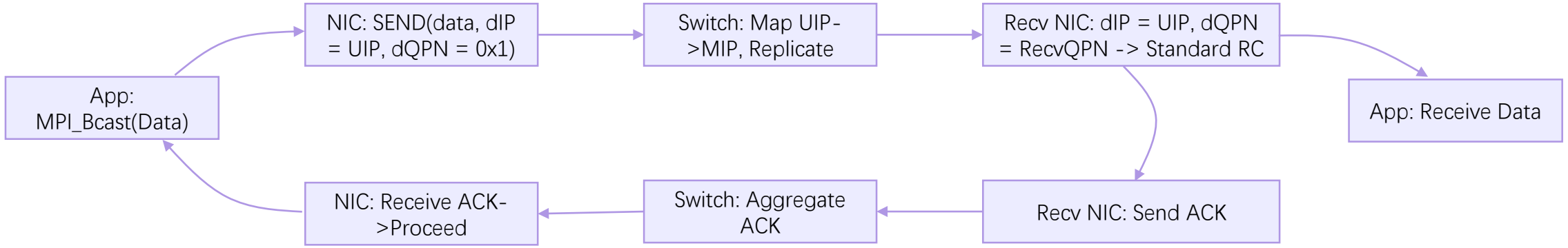
Solution	Reliability	Scalability	Issue
Application-layer unicast emulation	☑	✕	Sender replicates N times, bandwidth/CPU scales linearly
Native RDMA multicast (UD mode)	✕	☑	Unreliable, unordered, PMTU-limited, unsuitable for MPI

Core Conflict: The network can replicate, but applications dare not use it.

Group RDMA's Goal: Achieve **true reliable, efficient, and scalable multicast** over commercial RDMA networks.

Architecture: End-Net Collaboration, Virtual Meets Real

GroupRDMA adopts an "end-standard, net-intelligent" architecture, using the switch as a "protocol proxy" to present a "reliable multicast RC connection" to applications.



End-side: Fully compatible with standard RDMA Verbs, no app/NIC changes.

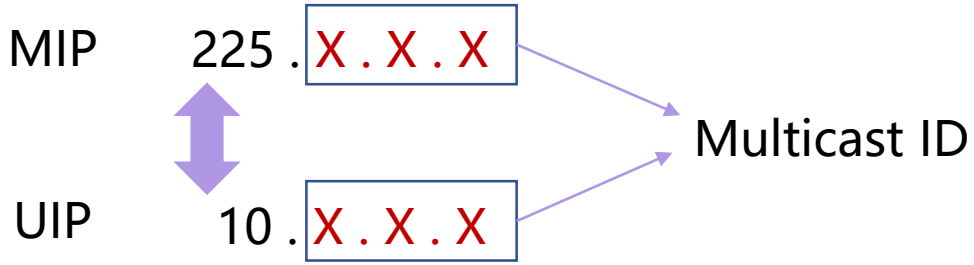
Net-side: Switch handles address proxy, packet replication, ACK aggregation, path management.

Key Technology 1: Hybrid Proxy IP & Virtual QP Model

Problem: Commercial NICs often reject RC connections with multicast destination IP

Solution: Unicast Proxy IP (UIP) + Multicast IP (MIP) + Virtual QP

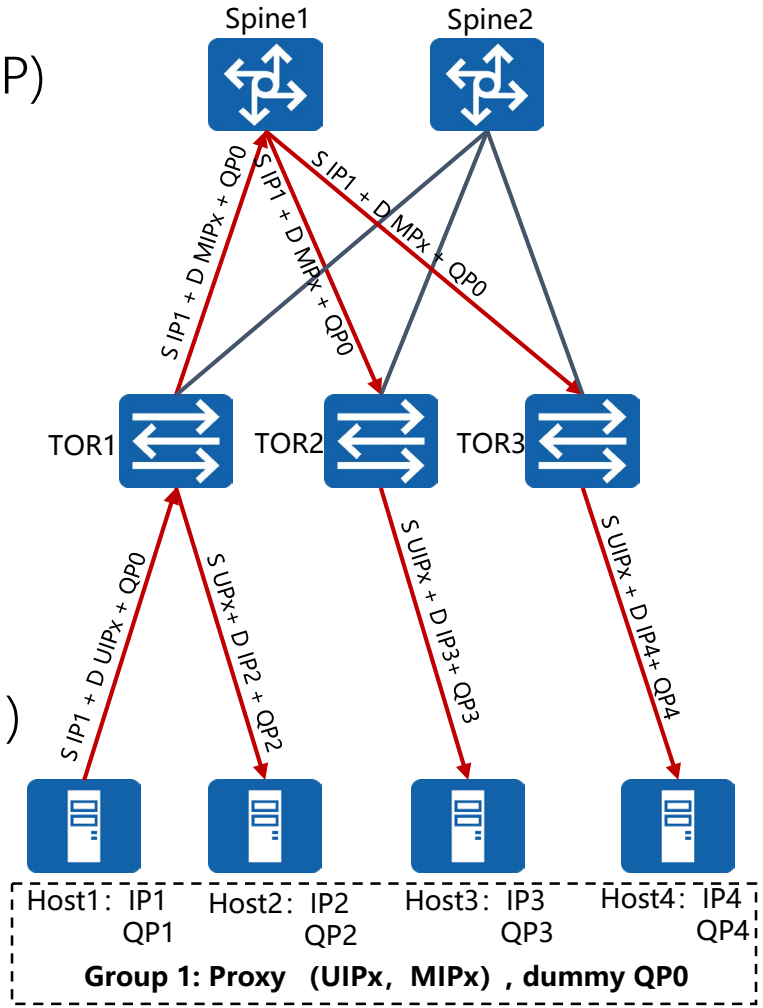
- End-side Connection: All nodes establish RC QP to a Virtual QP at a Unicast DIP (UIP), a logical group identifier
  - e.g., dIP = 10.0.0.10, dQPN = 0x1
- Net-side Mapping: Switch maintains UIP → MIP mapping, proxying unicast into multicast
  - e.g., UIP : 10.0.0.10 -> MIP : 225.0.0.10



Advantage: Bypasses NIC compatibility issues while preserving multicast scalability

Key Technology 1: Forward Data Path

- Sender NIC sends: srcIP=SrcIP, dstIP=UIP, dstQPN=0x1(VirtualQP)
- ToR switch:
  - Recognizes UIP as a multicast group
  - Maps UIP → MIP (e.g., 225.0.0.10)
  - Changes dstIP to MIP, enters multicast forwarding
- Spine/ToR replicates and forwards based on MIP
- Receiving ToR:
  - Maps MIP → UIP (per-receiver mapping)
  - Rewrites dstIP=UIP, dstQPN=RecvQPN (actual receiver QPN)
- Receiver NIC processes as standard unicast RC
- Key Point: End sees "unicast RC", network performs "multicast"



Key Technology 2: ACK Aggregation (End-to-End Reliability)

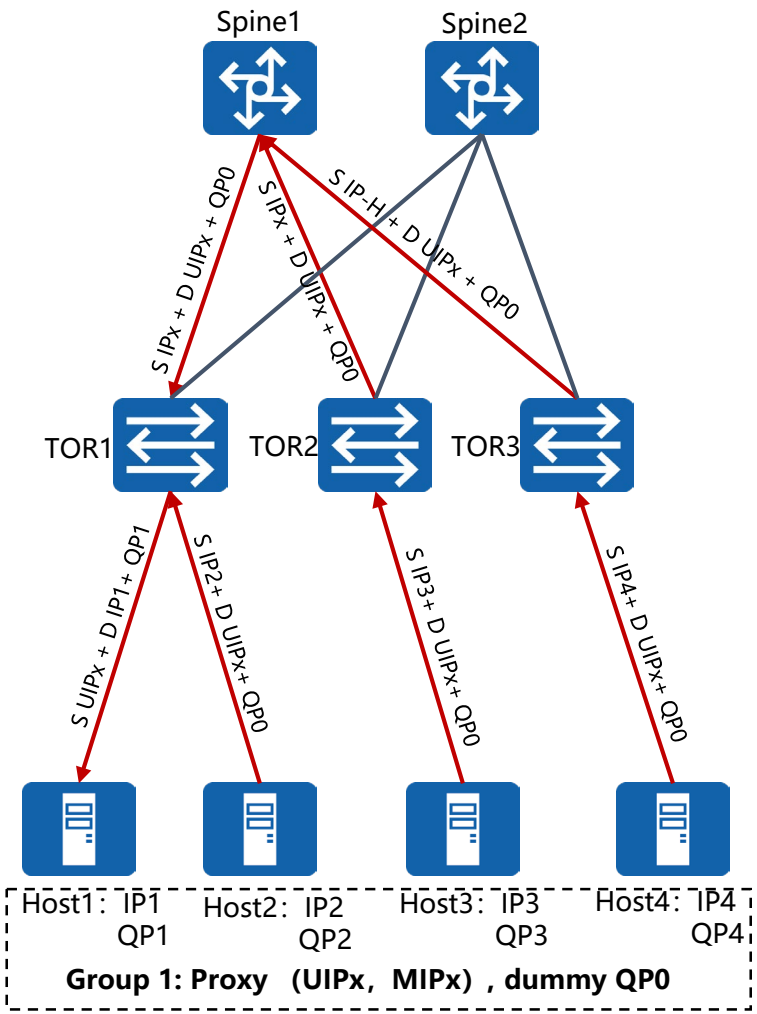
Problem: How to satisfy the sender?

- Standard RC requires ACKs. Forwarding all ACKs violates RC semantics.

Solution: Intelligent ACK Aggregation in Switch

- Switch performs ACK Aggregation per multicast group
- Aggregation Rule: Only forward ACK(PSN=X) when all receivers have ACKed PSN=X

Result: Sender sees a continuous, reliable, RC-compliant ACK stream



Key Technology 3: BIDI-PIM, Built for Massive Scale

Problem: Conventional PIM Faces Table Explosion in Large-Scale Scenarios

- In HPC/AI clusters, a single task may involve thousands of nodes in multicast, and each node can act as a sender, traditional PIM-SM:
  - Each (Source, Group) pair requires a separate (S, G) multicast routing entry
  - For a scenario with N senders and M multicast groups, the number of multicast entries scales to N × M
  - Core Conflict: Business demand (massive multicast) vs. hardware constraint (limited table entries)

Solution: Source-Agnostic BIDI-PIM

- Bidirectional RPT based solely on (G) entries, Regardless of the number of sources
- Effect: Reduces multicast entries from O(N×M) to O(M), achieving an order-of-magnitude compression.

Key Technology 4: Control Plane – Group Establishment

Problem: How to enable end-to-end automation

Solution: A lightweight, UDP-based control protocol that carries group metadata

Field	Purpose
opCode	Operation: CREATE, SWITCHOVER, RELEASE
# of Node	Number of receivers
Node List	(IP, QPN) pairs

- MIP Allocation: Rank0 requests an MIP from a designated swtich
- Membership Discovery: Rank0 collects all receiver IPs and QPNs via MPI runtime
- Envelop Dispatch: Rank0 sends one CREATE message to UIP, All nodes receive it; switches parse and configure
- Switch Configuration: Build UIP <-> MIP mapping, Join BIDI-PIM (*, G) tree, Initialize ACK aggregation state

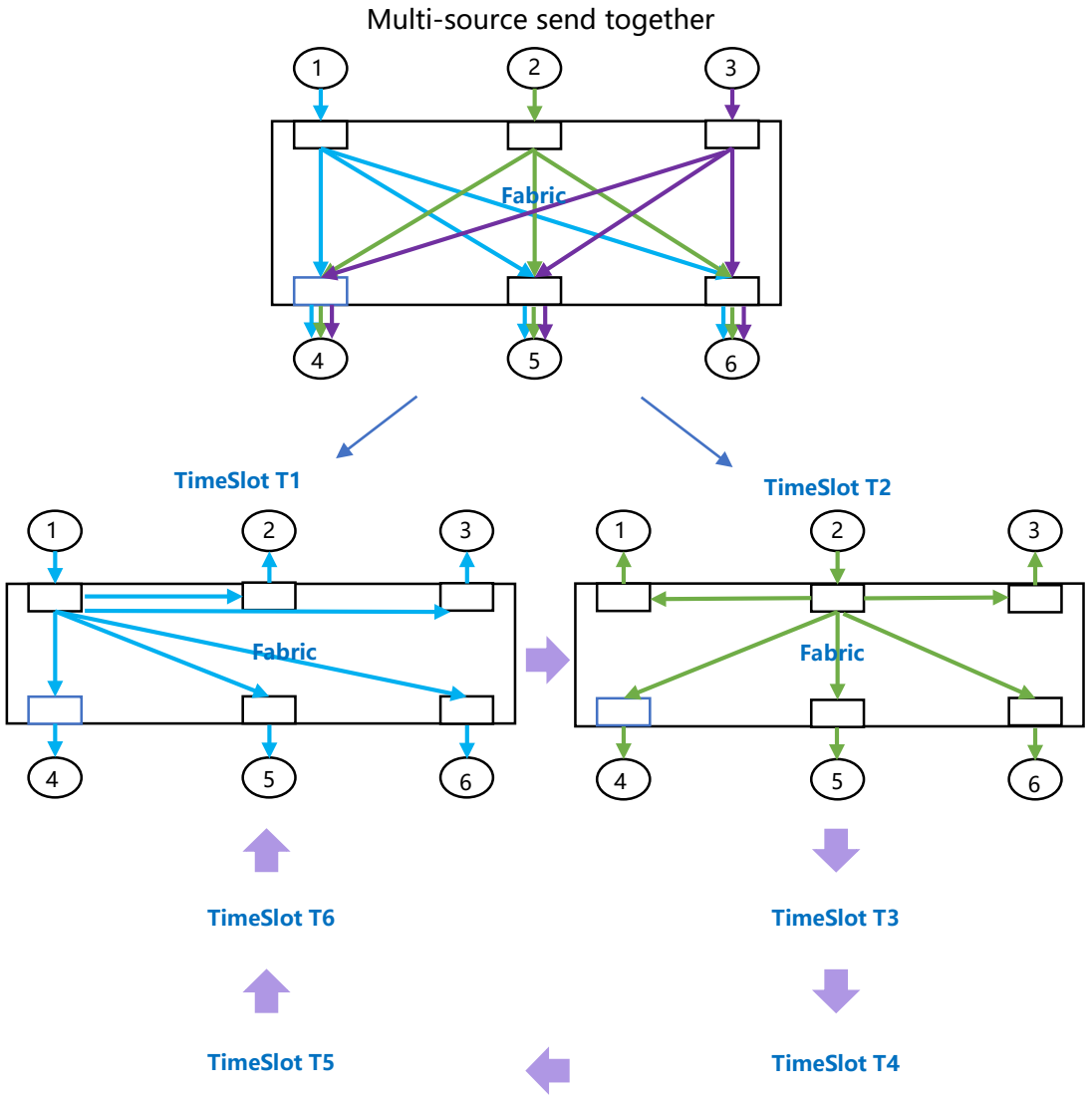
Key Technology 5: Multiple Source Switchover

Problem: Multiple senders may cause conflicts and confusion

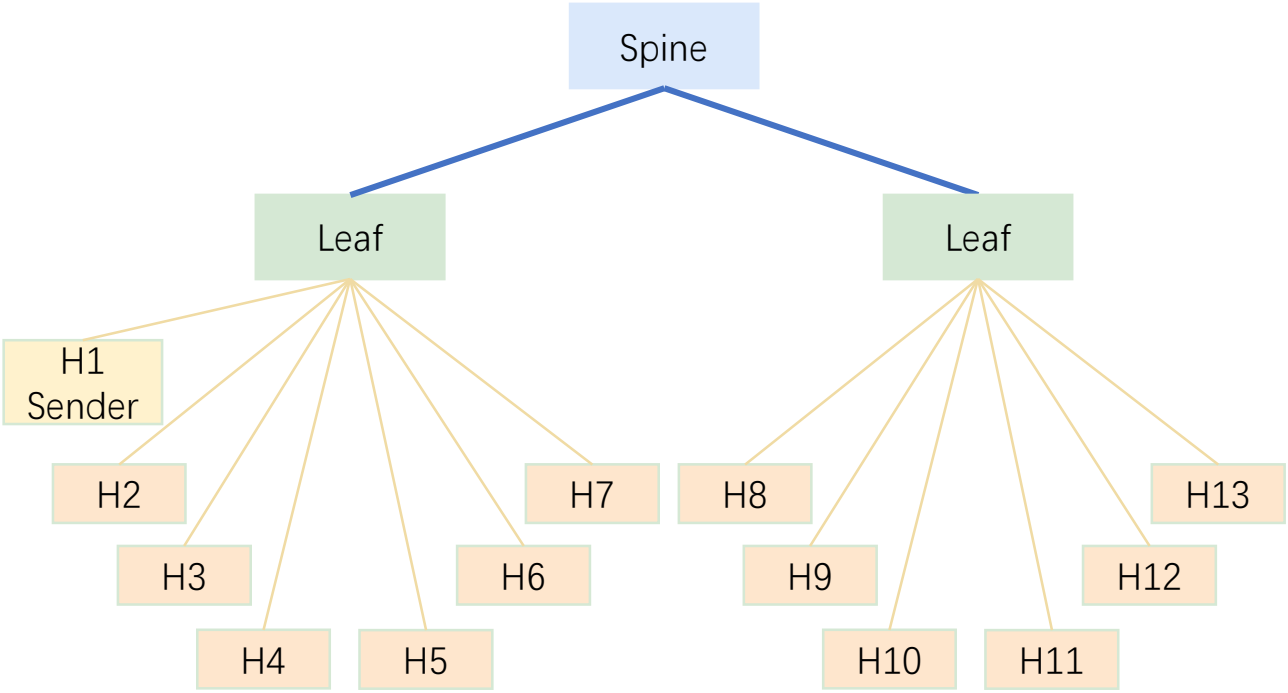
- Simultaneous packet transmission from multiple sources may cause congestion and the receiver cannot distinguish between these senders
  - All senders share one virtual QP

Solution: Timeslot based Multiple Source Switchover

- Only one sender active per timeslot.
- Use switch message to notify other senders to switch source

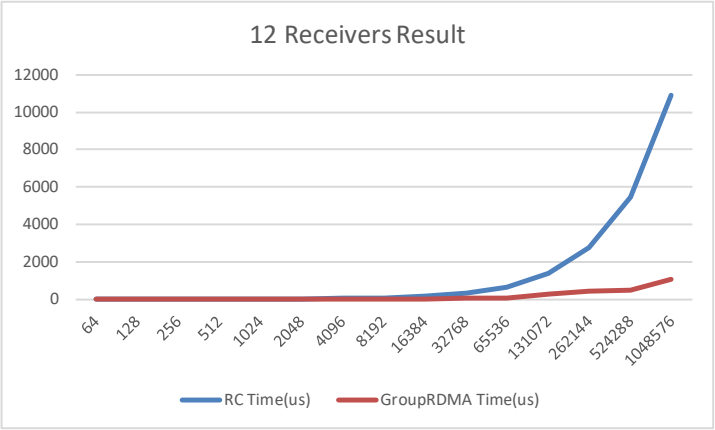


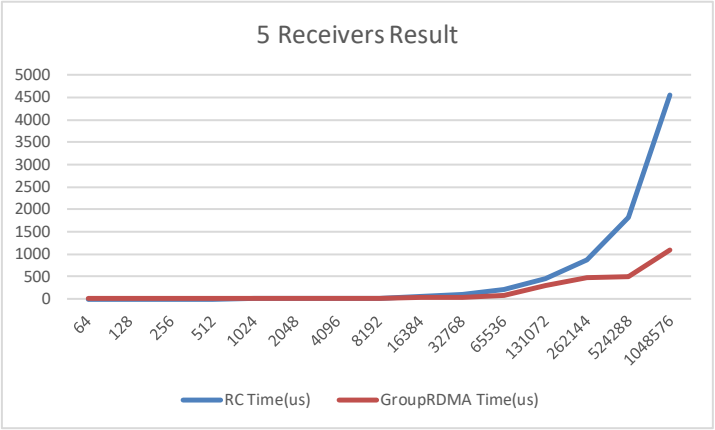
Test Methodology for Group RDMA

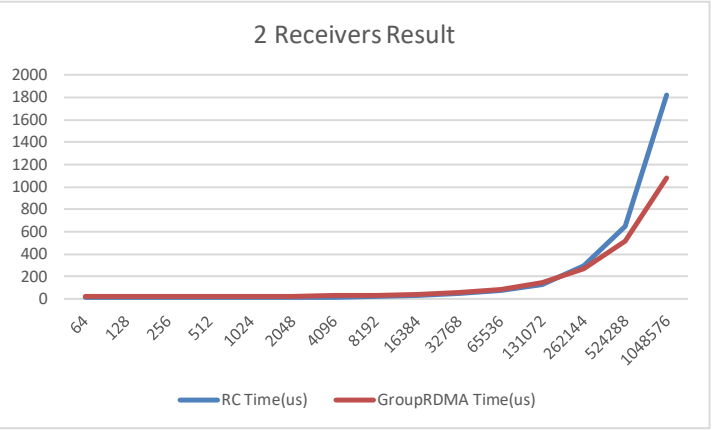


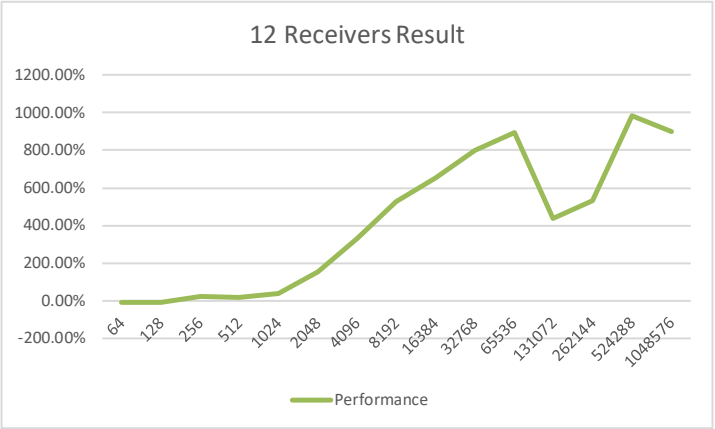
- Operation: MPI_Bcast with variable payload sizes
- Metric: Avg. end-to-end completion time (over 1,000 runs)
- Baseline : RC-based linear algorithm

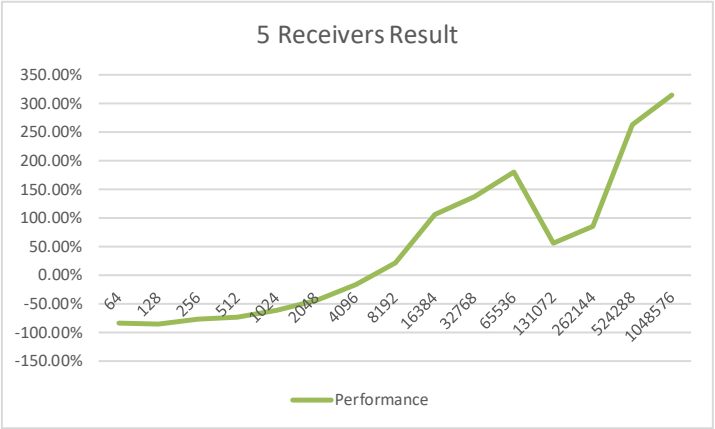
Performance: Near-Linear Scaling

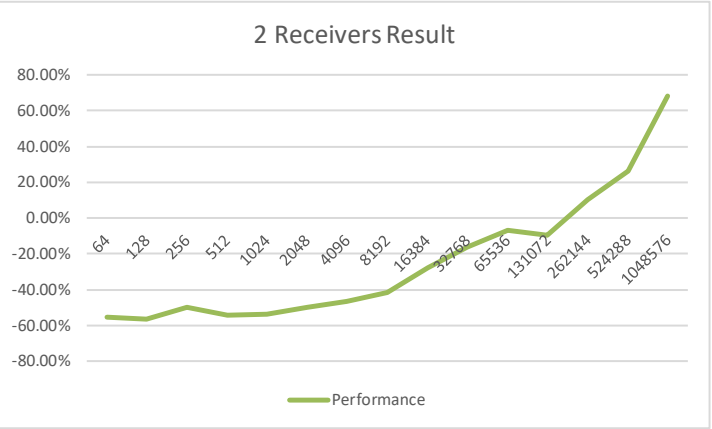












- 12 receivers and messages larger than 256 bytes, enabling up to 10x speedup
- Particularly efficient in HPC and AI computing workloads

Conclusion

GroupRDMA, through

Hybrid Proxy IP + **Virtual QP** + **ACK Aggregation** + **BIDI-PIM** + **DCT Switchover**,

delivers high, reliable RDMA multicast on commercial hardware,
without requiring changes to applications or NICs.

— a revolutionary solution for HPC/AI clusters